PREPRINT

Calibrated Critics: Learning When to Trust Rubric Rewards in Open-Ended RL

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ABSTRACT

Rubric-based reinforcement learning has become a standard way to post-train language models on tasks without verifiable answers, yet the judge that scores each rubric item is itself an imperfect proxy. We show that the judge's confidence is informative: rubric items on which the judge is uncertain account for most of the reward that a policy later exploits. We introduce *Calibrated Critics*, which reweights each criterion by an estimate of the judge's reliability on that criterion and prompt. Across long-form research and medical benchmarks, our method improves held-out gold judge scores by up to 6.1 points while matching the baseline's in-domain reward, and it substantially narrows the gap between training and gold judges throughout training.

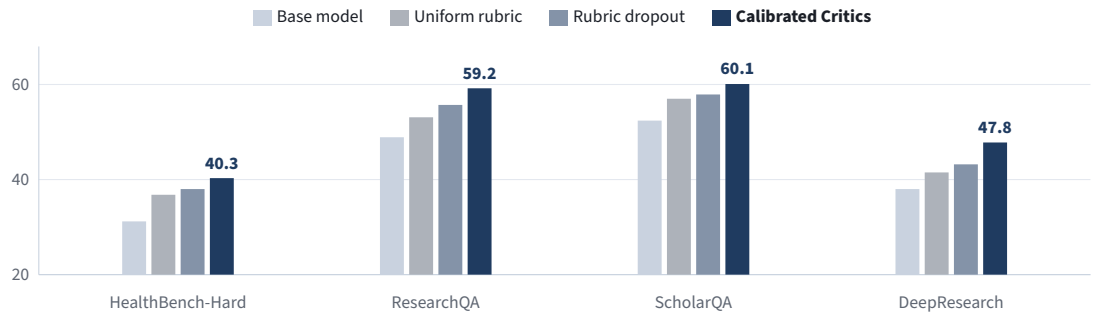


Figure 1 **Out-of-distribution results.** Gold judge scores on four long-form benchmarks. Calibrated Critics improves over every baseline on all four.

1 Introduction

Reinforcement learning with verifiable rewards works well when a ground truth exists (Shao et al., 2024). Many of the tasks we care most about, such as explaining a research area or giving medical

Preprint, work in progress. Code and checkpoints will be released on the project page.

advice, are open-ended. A common recipe is to write a rubric for each prompt and let an LLM judge grade each criterion (Gunjal et al., 2025). The rubric, however, is a proxy, and a policy optimized against it long enough learns to exploit the difference (Gao et al., 2023).

Our contributions are as follows:

- We show that judge uncertainty predicts which rubric items will be exploited, before exploitation happens.
- We propose a per-criterion reliability weight that is cheap to estimate and drops into any GRPO-style objective.
- We evaluate on four long-form benchmarks and two model scales.

TAKEAWAY 1

Judge uncertainty identifies the rubric criteria most vulnerable to reward hacking. Most exploitation concentrates on a small set of criteria where the judge was already uncertain before policy optimization.

2 Method

2.1 Reliability-weighted rubric reward

Given a prompt x with rubric $\{c_k\}_{k=1}^{K}$ and weights w_k, the standard reward for a response y is the weighted fraction of satisfied criteria. We instead use

$$r(x, y) \;=\; \frac{\sum_{k=1}^{K} w_k \, \rho_k(x) \, \mathbb{I}[\, y \models c_k \,]}{\sum_{k=1}^{K} w_k \, \rho_k(x)}, \tag{1}$$

where $\rho_k(x) \in [0, 1]$ is the estimated reliability of the judge on criterion c_k.

OBSERVATION 1

Judge disagreement is concentrated. A small fraction of criteria accounts for most of the judge's vote entropy, and these are predominantly stylistic rather than factual criteria.

Estimating reliability. We sample the judge m times per criterion and set ρ_k to one minus the normalized entropy of its votes.

DISCUSSION 1

Relationship to rubric dropout. Dropout makes a fixed rubric harder to exploit by varying which criteria receive reward. Reliability weighting addresses a complementary failure mode: it reduces the influence of criteria whose judgments are unstable even before the policy begins to optimize against them.

3 Experiments

Setup. We post-train two policy scales with GRPO against three baselines: the base model, the uniform rubric reward, and rubric dropout, which masks a random subset of criteria at every step. We evaluate on four held-out benchmarks:[1]

[1] All benchmarks are scored by a gold judge that is never used during training; Appendix A.2 gives its configuration.

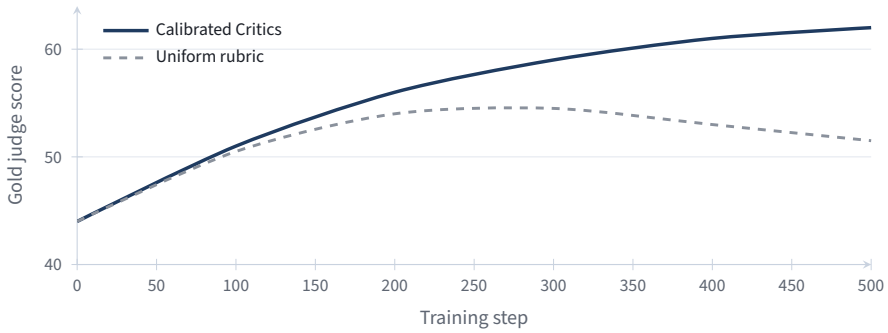


Figure 2 **Training dynamics.** Gold judge score on held-out prompts during training. The uniform rubric baseline peaks and then declines as the policy exploits unreliable criteria; reliability weighting keeps improving.

Table 1 **Main results.** Gold judge scores on out-of-distribution benchmarks. Best values are shown in bold.

Method	HealthBench-Hard	ResearchQA	ScholarQA	DeepResearch	Avg.
Base model	31.2	48.9	52.4	38.0	42.6
Uniform rubric	36.8	53.1	57.0	41.5	47.1
Rubric dropout	38.0	55.7	57.9	43.2	48.7
Calibrated Critics	**40.3**	**59.2**	**60.1**	**47.8**	**51.9**

1. **HealthBench-Hard**: open-ended clinical advice graded against physician-written rubrics.
2. **ResearchQA** and **ScholarQA**: literature-synthesis questions with citation-level criteria.
3. **DeepResearch**: multi-document report writing.

3.1 Main results

Table 1 reports gold judge scores on all four benchmarks and Figure 2 shows the training dynamics. The gain is largest on DeepResearch, whose rubrics contain the most criteria the judge is uncertain about, and smallest on ScholarQA, whose criteria are mostly citation checks that the judge grades consistently.

MAIN FINDING

Reliability weighting improves gold judge scores on every benchmark while leaving in-domain rubric reward unchanged, consistent with Takeaway 1.

3.2 How many judge samples are needed?

Figure 3 varies the number of judge samples m used to estimate ρ_k. Most of the benefit arrives by $m = 3$; beyond $m = 5$ the estimate is stable and the extra judge calls are wasted.

PRACTICAL GUIDANCE 1

Calibrate the judge before starting reinforcement learning. Sample each rubric criterion several times on representative prompts, estimate its agreement, and downweight unstable criteria. Recompute these estimates periodically as the policy distribution shifts.

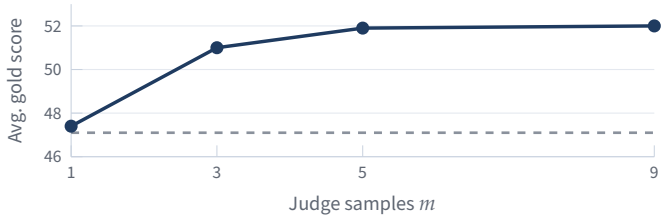


Figure 3 **Judge-sample ablation.** Average gold score as a function of m; the dashed line is the uniform rubric baseline.

4 Limitations

Reliability weighting assumes that judge disagreement signals a weak criterion rather than a hard response; when the two coincide, it down-weights exactly the criteria that matter most.

References

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.

Anisha Gunjal, Anthony Wang, Elaine Lau, et al. Rubrics as rewards: Reinforcement learning beyond verifiable domains. *arXiv preprint arXiv:2507.17746*, 2025.

Zhihong Shao, Peiyi Wang, Qihao Zhu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

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A Implementation details

A.1 Training hyperparameters

We train with GRPO using 8 rollouts per prompt, a learning rate of 10^{-6}, and a KL coefficient of 0.001 for 500 steps.

A.2 Judge configuration

The training judge is sampled $m = 5$ times per criterion at temperature 0.7.

B Prompts

RUBRIC JUDGE PROMPT

```
You are grading a response against a single rubric criterion.

Question: {question}
Response: {response}
Criterion: {criterion}

Answer YES if the response satisfies the criterion, otherwise NO.
```

The judge's output format is fixed with a second, machine-readable prompt, pasted verbatim:

JUDGE OUTPUT FORMAT

```
Question: {question}
Response: {response}
Criterion: {criterion}

Return JSON: {"verdict": "YES" | "NO", "confidence": 0-100}
# confidence is a percentage; use max_tokens=64.
```

C Additional results

Per-benchmark training curves are shown in the supplementary material.